THE AMERICAN COLLEGE, MADURAI

(An Autonomous Institution Affiliated to Madurai Kamaraj University)
Re-accredited (3rd Cycle) by NAAC with Grade "A+", GCPA -3.47 on a 4 point of scale

Since 1881

June Repeat Examination, 2025

Course Code: 24PBT4313 **Max Marks: 100**
Course Title: Introduction to Bioinformatics **Time: 3 Hrs**

PART – A

Answer all questions **(10 x 2 = 20)**

1. Identify the role of bioinformatics in personalized medicines.
2. What is the function of ribosomes.
3. Distinguish between primary and secondary database.
4. Define CATH.
5. What do you understand by global alignment?
6. List any two similarity search engine.
7. Write a short note on UPGMA
8. Define skewness test.
9. What is purpose of molecular docking?
10. Name any two bioinformatics tools used in target validation.

PART – B

Answer all questions (Either a or b) **(5 x 7 = 35)**

11. a. Elaborate on Human genome project.

(OR)

b. Explain the useful websites in Bioinformatics.

12. a. Discuss in detail about Primary databases

(OR)

b. Describe the purpose of metabolic pathway database.

13. a. Compare and contrast Global and local alignment.

(OR)

b. Explain the methods of multiple sequence alignment.

14. a. Comment on the distance-based methods.

(OR)

b. List out the steps involved in phylogenetic tree construction.

15. a. Discuss ligand-based approaches in drug discovery in detail.

(OR)

b. Explain the concept of ADMET.

PART – C

Answer any THREE questions **(3 x 15 = 45)**

16. Explain in detail about the scope of bioinformatics in various fields?
17. Discuss the importance of biological databases in bioinformatics.
18. Using the Needleman – Wunsch algorithm, align the two given DNA sequences
    Seq 1: AGCGAT & Seq 2: ACGAAT
19. Elaborate on the methods of constructing phylogenetic tree in detail.
20. Describe the role of bioinformatics in drug discovery.

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